

Welcome HackGT13 - 2026 class,

This is the HEARSAY challenge from NSA

Your mission should you choose to accept it is to defeat the Depepticodes ^

The Depepticodes have been perfecting synthetic speech from Text and cloning of voices. They can impersonate people with as little as 3 to 30 seconds of speech to use for stealing biometric identities and impersonating people. They've stolen millions of dollars in doing so. They have also been able to add varying levels of noise and in multiple languages. But here we're only dealing with English and clips greater than 3 seconds for the testing dataset.



We need your help in figuring out news ways to distinguish real from synthetic speech.

You will be given a real speech corpus known as LJ or LindaJohnson speech as well as a large dataset from Purdue called DiffSSD. The paths to these on HexLabs environment will be given to you in Discord Hearsay channel when that goes live. DiffSSD has data from a variety of Text to Speech generator outputs such as ElevenLabs and others including the latest Diffusion based models. Those datasets and any additional ones you might get are used for training. Any other datasets you might get from HuggingFace or other locations, that you are permitted to use can be added to your corpus of training.

For testing you will have a zip file of 1671 .wav files encoded at 16 KHz and an answer key to fill out with uniform answers. You have to modify the values. The answer key will be in tab separated values, .tsv format, analagous to .csv files. Each team will customize their copy of the answer key to reflect their team name. The 2nd column is the score that ranges from 0.0 for 100% confident Real to 1.0 for 100% confident synthetic.

Approximately 70% of the files are real.

We're taking the Analyst scenario so we are running our detection for the purpose of providing results to Intelligence Analysts. Every False Alarm, identifying a real audio as fake takes up analyst's time and causes a loss of confidence in the algorithm. This is no balanced accuracy scenario. So we are using the Detection Cost Function of the ASVSpooF v5 challenge (Automatic Speaker Verification) track 1. Where there will be a penalty of 4 times greater for a false positive aka false alarm vs a missed detection.

Other scenarios such as filtering out deep fakes for training might have a completely different penalty scenario. In real situations, this will often be a factor.

You will have to strategize and use some sort of Game Theory or Benchmark maxing to determine what the default answer to be if you cannot determine an answer for all of the files. This could be crucial!

In order to keep track of files that you have done and not done there is a default value of .006 which has no significance, it is just not 0.0. Do not remove any rows as they will be needed to compare scores.

Overall challenge evaluation will be 60% for the minDCF score. 20% for creativity, quality, innovation and depth and 20% for documentation from a GitHub repository. The best minDCF score is 0.0 and the worst is 1.0. The best, lowest minDCF will get 60 points. The rest will get the percentage of the best DCF multiplied by 60. So if the best minDCF is .1 and the next best is .2. The second team will get $(1.0-.2)/(1.0-.1) * 60$ or 53.3 points. If we decide that this algorithm is unworkable we will update you.

Prizes as outlined in slides consist of Sonos Alexa Speakers for first place, Audio-Technica USB Mic and JBL Bluetooth Speakers for each team member.

Good luck, be sure to ask questions in the Discord channel, look for the HearSay tag.. Any questions you ask personally will be broadcast to all of the groups if deemed applicable to all.

These instructions will not! Self Destruct in 5 seconds.

References – More to come in Discord!

ASVSpooof GitHub <https://github.com/asvspoof-challenge/asvspoof5> (Track 1! For MinDCF, also at HexLabs)

ASVSpooof Arxiv paper <https://arxiv.org/html/2502.08857v3>

DiffSSD <https://arxiv.org/abs/2409.13049>

LJ Speech <https://keithito.com/LJ-Speech-Dataset/>

<https://huggingface.co/datasets?modality=modality:audio&p=97&sort=trending> Hugging Face Audio Datasets 100 pages!